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SE LAB 3: ARCHITECTURE SELECTION AND JUSTIFICATION

System: Book-Exchange Platform

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Architecture Selection: Layered Architecture

We chose Layered Architecture for the Book-Exchange Platform, a peer-to-peer marketplace where students list books to lend or borrow and coordinate exchanges through matched requests. The architecture separates the user interface, application services, and relational data storage.

Reasons for Selection

1. Clear separation of responsibilities: The user interface handles book discovery and user actions, the application services manage authentication, listing CRUD operations and exchange matching, and the data layer stores users, listings and request statuses. This structure makes the system easier to maintain and test.

2. Suitable for relational data and matching: Book listings, student accounts and exchange requests have relationships that can be represented in a relational database. A dedicated matching service can query listings and coordinate compatible requests without mixing matching logic into the interface.

Security Advantage

The Authentication and User Service can verify identity and enforce authorization before allowing users to create, edit or delete listings or submit exchange requests. Passwords should be stored as secure hashes, and database access should use parameterized queries and least-privilege credentials.

Performance Benefit

Separating listing search and matching logic from the interface allows database indexes on fields such as book title, author, availability and request status. These indexes can reduce lookup time as the number of listings and exchange requests grows.

Conclusion

Layered Architecture provides a practical and maintainable design for the platform's authentication, book-listing CRUD operations, matching workflow and relational database.